Population Attributable Fraction of Obesity for Hypercapnic Respiratory Failure

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1 Obesity and hypercapnic respiratory failure analysis

1.1 Section 0. Setup

1.1.1 Load packages

```
library(haven)
library(ggplot2)
library(modelsummary)
```

```
Warning: package 'modelsummary' was built under R version 4.5.3
```

```
library(here)
```

```
Warning: package 'here' was built under R version 4.5.3
```

```
here() starts at C:/Users/ajsch/Downloads
```

```
library(gtsummary)
library(stringr)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

1.1.2 Define file paths and constants

```r
data_path <- here::here("full_db.rds")

co2_abg_cutoff <- 45
co2_vbg_cutoff <- 50
ph_cutoff <- 7.35
icd_hrf_vars <- c("has_j9612", "has_j9622", "has_j9692", "has_j9602")

coef_map_common <- c(
  "curr_bmi_num" = "BMI (per 1 kg/m^2)",
  "age_at_encounter" = "Age (per year)",
  "sexMale" = "Male",
  "ethnicityHispanic or Latino" = "Hispanic/Latino",
  "ethnicityUnknown" = "Ethnicity: Unknown"
)
```

1.2 Section 1. Data ingest and harmonization

1.2.1 readRDS()

```r
stata_data <- readRDS(data_path)
```

1.2.2 Standardize types (labelled -> numeric, factors)

```r
# Verify before conversion
table(stata_data$niv_proc, useNA = "ifany")
```

```
                                  No NIV
                                  799385
Received NIV (not including nightly CPAP)
                                   34091
```

```r
# Convert labelled variables to numeric properly
stata_data <- stata_data %>%
  mutate(
    niv_proc = case_when(
      as.character(as_factor(niv_proc)) == "Received NIV (not including nightly CPAP)" ~ 1,
      as.character(as_factor(niv_proc)) == "No NIV" ~ 0,
      TRUE ~ NA_real_
    ),
    imv_proc = case_when(
      as.character(as_factor(imv_proc)) == "Received IMV" ~ 1,
      as.character(as_factor(imv_proc)) == "No IMV" ~ 0,
      TRUE ~ NA_real_
    ),
    highest_paco2 = as.numeric(highest_paco2),
    abg_ph = as.numeric(abg_ph),
    vbg_co2 = as.numeric(vbg_co2),
    vbg_ph = as.numeric(vbg_ph)
  )
```

```r
# Verify after conversion
table(stata_data$niv_proc, useNA = "ifany")
```

```
     0      1
799385  34091
```

```r
stopifnot(all(is.na(stata_data$niv_proc) | stata_data$niv_proc %in% c(0, 1)))
stopifnot(any(stata_data$niv_proc == 0, na.rm = TRUE))
stopifnot(any(stata_data$niv_proc == 1, na.rm = TRUE))

stopifnot(all(is.na(stata_data$imv_proc) | stata_data$imv_proc %in% c(0, 1)))
stopifnot(any(stata_data$imv_proc == 0, na.rm = TRUE))
stopifnot(any(stata_data$imv_proc == 1, na.rm = TRUE))

any_vent <- (stata_data$niv_proc == 1 | stata_data$imv_proc == 1)
stopifnot(any(any_vent == TRUE, na.rm = TRUE))
stopifnot(any(any_vent == FALSE, na.rm = TRUE))
```

1.2.3 Quick sanity checks (row counts, missingness)

```r
nrow(stata_data)
```

```
[1] 833476
```

TODO: Add missingness checks.

```r
library(dplyr)

stata_data %>%
  summarise(
    total_encounters = n(),
```

```
    niv_count = sum(niv_proc == 1, na.rm = TRUE),
    imv_count = sum(imv_proc == 1, na.rm = TRUE),
    any_vent = sum(niv_proc == 1 | imv_proc == 1, na.rm = TRUE),
    neither = sum(niv_proc == 0 & imv_proc == 0, na.rm = TRUE)
  )
```

total_encounters	niv_count	imv_count	any_vent	neither
833476	34091	58232	85678	747798

BMI missingness:

```
table(ifelse(is.na(stata_data$curr_bmi), "BMI Missing (NA)", "BMI Value Present"))
```

```
 BMI Missing (NA) BMI Value Present
           450715             382761
```

VBG, ABG, and CO2 missingness:

```
library(dplyr)
library(tidyr)

missing_summary <- tibble(
  variable = c("VBG CO2", "Highest PaCO2", "ABG pH", "VBG pH"),
  missing = c(
    sum(is.na(stata_data$vbg_co2)),
    sum(is.na(stata_data$highest_paco2)),
    sum(is.na(stata_data$abg_ph)),
    sum(is.na(stata_data$vbg_ph))
  ),
  present = c(
    sum(!is.na(stata_data$vbg_co2)),
    sum(!is.na(stata_data$highest_paco2)),
```

```
    sum(!is.na(stata_data$abg_ph)),
    sum(!is.na(stata_data$vbg_ph))
  )
) %>%
  mutate(
    total = missing + present,
    pct_missing = round(100 * missing / total, 1)
  )

missing_summary
```

variable	missing	present	total	pct_missing
VBG CO2	677009	156467	833476	81.2
Highest PaCO2	635606	197870	833476	76.3
ABG pH	633941	199535	833476	76.1
VBG pH	666142	167334	833476	79.9

```
table(
  "VBG CO2 & VBG pH availability" =
    ifelse(
      is.na(stata_data$vbg_co2) | is.na(stata_data$vbg_ph),
      "One or Both Missing",
      "Both Present"
    )
)
```

```
VBG CO2 & VBG pH availability
       Both Present One or Both Missing
             156399              677077
```

```
table(
  "Highest PaCO2 & ABG pH availability" =
    ifelse(
```

```
      is.na(stata_data$highest_paco2) | is.na(stata_data$abg_ph),
      "One or Both Missing",
      "Both Present"
    )
)
```

```
Highest PaCO2 & ABG pH availability
       Both Present One or Both Missing
             197141              636335
```

```
stata_data %>%
  summarise(
    bmi_min = min(curr_bmi, na.rm = TRUE),
    bmi_max = max(curr_bmi, na.rm = TRUE),
    paco2_min = min(highest_paco2, na.rm = TRUE),
    paco2_max = max(highest_paco2, na.rm = TRUE),
    ph_min = min(abg_ph, na.rm = TRUE),
    ph_max = max(abg_ph, na.rm = TRUE)
  )
```

bmi_min	bmi_max	paco2_min	paco2_max	ph_min	ph_max
10.09	50	6	249	5.912479	8.011059

are things missing more in one group vs another? would need to put this downstream because the groups aren't defined yet

BMI missingness non-HRF vs HRF

BMI missingness between NIV and IMV

ABG pH missingness between NIV and IMV

VBG pH missingness between NIV and IMV

1.3 Section 2. Variable derivations

1.3.1 BMI parsing + BMI categories

```
bmi_raw <- stata_data %>%
  transmute(curr_bmi_raw = na_if(str_trim(as.character(zap_labels(curr_bmi))), "n/a"))

non_numeric_bmi <- bmi_raw %>%
  filter(!is.na(curr_bmi_raw), !str_detect(curr_bmi_raw, "^\\d+(\\.\\d+)?$"))

non_numeric_bmi %>%
  count(curr_bmi_raw, sort = TRUE)
```

curr_bmi_raw	n

```
non_numeric_bmi %>%
  summarise(n_failed_parse = n())
```

n_failed_parse
0

```
stata_data <- stata_data %>%
  mutate(
    curr_bmi_num = if_else(
      str_detect(str_trim(as.character(zap_labels(curr_bmi))), "^\\d+(\\.\\d+)?$"),
      as.numeric(str_trim(as.character(zap_labels(curr_bmi)))),
      NA_real_
    )
  )
```

```r
stata_data <- stata_data %>%
  mutate(
    bmi_category = case_when(
      is.na(curr_bmi_num) ~ NA_character_,
      curr_bmi_num < 18.5 ~ "Underweight",
      curr_bmi_num < 25   ~ "Normal weight",
      curr_bmi_num < 30   ~ "Overweight",
      curr_bmi_num < 35   ~ "Obesity I",
      curr_bmi_num < 40   ~ "Obesity II",
      curr_bmi_num >= 40  ~ "Obesity III"
    )
  )
```

1.3.2 HRF definition v1 (gases + ventilation)

Below is my first (probably most stringent) iteration of hypercapnic respiratory failure: co2 greater than 45 if abg, 50 if vbg, ph <7.35 - they have to have either an abg or a vbg showing this, AND either NIV or IMV or both. I've written it so that ABG values and VBG values cannot mix

```r
stata_data <- stata_data %>%
  mutate(
    gas_source = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) ~ "ABG",
      TRUE ~ "VBG"
    )
  )
```

```r
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic = case_when(
      # ABG-based HRF
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "ABG" &
        highest_paco2 >= co2_abg_cutoff &
        abg_ph <= ph_cutoff ~ 1,
```

```
      # VBG-based HRF (only when ABG incomplete)
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "VBG" &
        !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= co2_vbg_cutoff &
        vbg_ph <= ph_cutoff ~ 1,

      TRUE ~ 0
    ),
    hrf_hypercapnic = factor(
      hrf_hypercapnic,
      levels = c(0, 1),
      labels = c("No", "Yes")
    )
  )
```

1.3.3 HRF definition v2 (labs + ICD) and v2_strict

Regression done with new definition of hypercapnic respiratory failure - pH and CO2 definition as above, except now including (required) ICD codes instead of ventilation (I only used 4 - the first four that are listed in database):

```
stata_data <- stata_data %>%
  mutate(
    # Lab-based hypercapnic respiratory failure (ABG or VBG)
    hrf_hypercapnic_labs_v2 = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) &
       highest_paco2 >= co2_abg_cutoff & abg_ph < ph_cutoff ~ 1, # BL note - fixed this 50->45

      !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= co2_vbg_cutoff & vbg_ph < ph_cutoff ~ 1,

      TRUE ~ 0
    ),

    # ICD-based hypercapnic respiratory failure
```

```
    hrf_hypercapnic_icd_v2 = ifelse(
      rowSums(across(all_of(icd_hrf_vars)) == 1, na.rm = TRUE) > 0,
      1, 0
    ),

    # Final combined HRF definition (labs OR ICD)
    hrf_hypercapnic_v2 = ifelse(
      hrf_hypercapnic_labs_v2 == 1 |
      hrf_hypercapnic_icd_v2 == 1,
      1, 0
    )
  )
```

```
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic_v2_strict = ifelse(
      hrf_hypercapnic_labs_v2 == 1 &
      hrf_hypercapnic_icd_v2  == 1,
      1, 0
    )
  )
```

1.4 Section 3. Cohort construction and attrition

1.4.1 Primary analysis cohort (HRF v1)

```
stata_hrf_analysis <- stata_data %>%
  filter(!is.na(curr_bmi_num),
         (!is.na(highest_paco2) & !is.na(abg_ph)) |
         (!is.na(vbg_co2) & !is.na(vbg_ph)))
```

1.4.2 Strict v2 cohort (sensitivity)

```
stata_hrf_strict_analysis <- stata_data %>%
  mutate(
    has_gas = (!is.na(highest_paco2) & !is.na(abg_ph)) |
              (!is.na(vbg_co2) & !is.na(vbg_ph))
  ) %>%
  filter(
    # Keep strict HRF cases
    hrf_hypercapnic_v2_strict == 1 |

    # OR clean controls with labs and no HRF by any definition
    (
      has_gas == TRUE &
      hrf_hypercapnic_labs_v2 == 0 &
      hrf_hypercapnic_icd_v2  == 0
    )
  )
```

TODO: Add attrition table with inclusion/exclusion counts.

```
attrition_primary <- tibble(
  step = c(
    "Initial dataset",
    "Exclude missing BMI",
    "Exclude missing ABG/VBG gases",
    "Final primary cohort"
  ),
  n = c(
    nrow(stata_data),
    sum(stata_data$has_bmi),
    sum(stata_data$has_bmi & stata_data$has_any_gas),
    nrow(stata_hrf_analysis)
  )
)
```

```
Warning: Unknown or uninitialised column: `has_bmi`.
Unknown or uninitialised column: `has_bmi`.
```

```
Warning: Unknown or uninitialised column: `has_any_gas`.
```

```
attrition_primary
```

step	n
Initial dataset	833476
Exclude missing BMI	0
Exclude missing ABG/VBG gases	0
Final primary cohort	112009

1.5 Section 4. Descriptive statistics

1.5.1 Distribution of BMI in the analytic cohort

```
table(stata_data$hrf_hypercapnic, useNA = "ifany")
```

```
    No    Yes
807702  25774
```

```
summary(stata_data$curr_bmi_num)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  10.09   26.31   33.34   33.64   41.20   50.00  450715
```

```
table(stata_data$bmi_category, useNA = "ifany")
```

```
Normal weight      Obesity I     Obesity II    Obesity III     Overweight
        65912          58317          44022         130477          74499
  Underweight           <NA>
         9534         450715
```

[BL - note high rates of BMI missing data]

1.5.2 HRF prevalence by BMI category (v1)

Counts (bottom right is amount of patients meeting criteria above, adds up to full dataset)

```
table(
  hrf = stata_data$hrf_hypercapnic,
  any_vent = (stata_data$niv_proc == 1 | stata_data$imv_proc == 1)
)
```

```
     any_vent
hrf    FALSE   TRUE
  No  747798  59904
  Yes      0  25774
```

1.5.3 Counts for both HRF as defined above AND current BMI above 30

```
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    curr_bmi_num >= 30
  ) %>%
  summarise(
    n_patients = n()
  )
```

n_patients
4099

Counts for HRF and BMI by category

```
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    !is.na(curr_bmi_num)
  ) %>%
  mutate(
    bmi_category = case_when(
      curr_bmi_num < 18.5 ~ "Underweight",
      curr_bmi_num < 25   ~ "Normal weight",
      curr_bmi_num < 30   ~ "Overweight",
      curr_bmi_num < 35   ~ "Obesity I",
      curr_bmi_num < 40   ~ "Obesity II",
      curr_bmi_num >= 40  ~ "Obesity III"
    )
  ) %>%
  count(bmi_category) %>%
  mutate(pct = round(100 * n / sum(n), 1))
```

bmi_category	n	pct
Normal weight	2311	25.1
Obesity I	1766	19.2
Obesity II	1131	12.3
Obesity III	1202	13.1
Overweight	2287	24.9
Underweight	503	5.5

1.5.4 Baseline characteristics table

☐ TODO

(still needs some work with the categories, cleaning up variables)

```
library(dplyr)
library(gtsummary)

tbl1 <- stata_data %>%
  mutate(
    # Outcome grouping
    hrf_group = case_when(
      hrf_hypercapnic == "Yes" ~ "HRF",
      hrf_hypercapnic == "No"  ~ "No HRF",
      TRUE ~ NA_character_
    ),

    # BMI grouping
    bmi_group = case_when(
      curr_bmi_num < 25 ~ "Normal",
      curr_bmi_num < 30 ~ "Overweight",
      curr_bmi_num >= 30 ~ "Obese",
      TRUE ~ "Missing BMI"
    )
  ) %>%
  select(
    hrf_group,
    bmi_group,
    curr_bmi_num,
    age_at_encounter,
    sex,
    ethnicity,
    highest_paco2,
    abg_ph,
    vbg_co2,
    vbg_ph,
    niv_proc,
    imv_proc
  ) %>%
```

```
  tbl_summary(
    by = hrf_group,
    missing = "ifany",
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    )
  ) %>%
  add_overall()

tbl1
```

1.5.5 Are proportions of hypercapnic respiratory failure (v1) different across BMI categories?

Just out of curiosity ran Chi square analysis below

```
bmi_hrf_subset <- stata_hrf_analysis %>%
  filter(!is.na(bmi_category))
```

```
bmi_hrf_summary <- bmi_hrf_subset %>%
  group_by(bmi_category) %>%
  summarise(
    n_total = n(),
    n_hrf = sum(hrf_hypercapnic == "Yes"),
    n_no_hrf = sum(hrf_hypercapnic == "No"),
    pct_hrf = round(100 * n_hrf / n_total, 1),
    pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
  ) %>%
  arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

chisq_test <- chisq.test(table(bmi_hrf_subset$bmi_category, bmi_hrf_subset$hrf_hypercapnic))

bmi_hrf_summary
```

Characteristic	**Overall** N = 833,476[1]	**HRF** N = 25,774[1]	**No HRF** N = 807,702[1]
bmi_group			
Missing BMI	450,715 (54%)	16,574 (64%)	434,141 (54%)
Normal	75,446 (9.1%)	2,814 (11%)	72,632 (9.0%)
Obese	232,816 (28%)	4,099 (16%)	228,717 (28%)
Overweight	74,499 (8.9%)	2,287 (8.9%)	72,212 (8.9%)
curr_bmi_num	34 (9)	30 (8)	34 (9)
Unknown	450,715	16,574	434,141
age_at_encounter	58 (17)	61 (16)	58 (18)
Sex			
Female	439,313 (53%)	10,763 (42%)	428,550 (53%)
Male	394,163 (47%)	15,011 (58%)	379,152 (47%)
Unknown or Other	0 (0%)	0 (0%)	0 (0%)
Ethnicity			
Not Hispanic or Latino	612,964 (74%)	20,459 (79%)	592,505 (73%)
Hispanic or Latino	55,675 (6.7%)	1,681 (6.5%)	53,994 (6.7%)
Unknown	164,837 (20%)	3,634 (14%)	161,203 (20%)
highest_paco2	49 (22)	70 (31)	46 (19)
Unknown	635,606	1,932	633,674
abg_ph	7.37 (0.11)	7.23 (0.11)	7.38 (0.10)
Unknown	633,941	1,826	632,115
vbg_co2	46 (13)	56 (19)	45 (11)
Unknown	677,009	13,857	663,152
vbg_ph	7.34 (0.11)	7.25 (0.14)	7.35 (0.10)
Unknown	666,142	12,988	653,154
niv_proc	34,091 (4.1%)	9,359 (36%)	24,732 (3.1%)
imv_proc	58,232 (7.0%)	19,105 (74%)	39,127 (4.8%)

[1] n (%); Mean (SD)

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	5061	503	4558	9.9	90.1
Normal weight	31060	2311	28749	7.4	92.6
Overweight	31756	2287	29469	7.2	92.8
Obesity I	21692	1766	19926	8.1	91.9
Obesity II	12223	1131	11092	9.3	90.7
Obesity III	10217	1202	9015	11.8	88.2

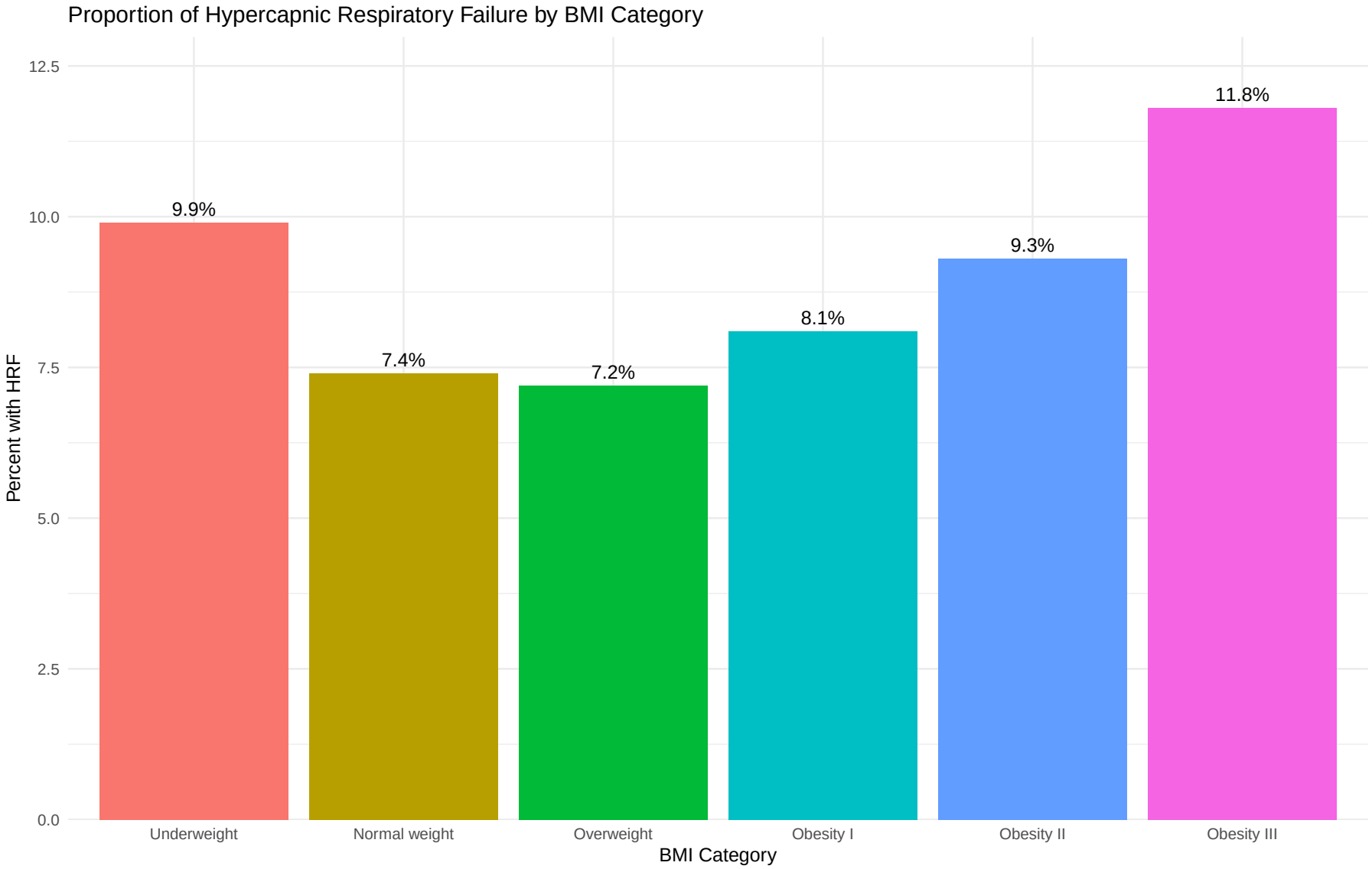
```
chisq_test$p.value
```

```
[1] 1.243203e-57
```

Visualization:

```
# Make sure bmi_category is ordered
bmi_hrf_summary <- bmi_hrf_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of HRF by BMI category
ggplot(bmi_hrf_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```


Proportion of Hypercapnic Respiratory Failure by BMI Category
Percent with HRF
12.5
10.0
7.5
5.0
2.5
0.0
9.9%
7.4%
7.2%
8.1%
9.3%
11.8%
Underweight
Normal weight
Overweight
Obesity I
Obesity II
Obesity III
BMI Category

1.6 Section 5. Primary inferential model

1.6.1 Unadjusted (aka simple) regression for outcome 1

Below is regression, starting with BMI (the degrees of freedom is 112k which seems appropriate - includes all patients with a BMI and labs/vent indicating NO HRF, plus the patients with BMI and hypercapnic respiratory failure as defined above)

```
logit_bmi_hrf <- glm(hrf_hypercapnic ~ curr_bmi_num,
data = stata_hrf_analysis,
family = binomial)

summary(logit_bmi_hrf)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num, family = binomial,
    data = stata_hrf_analysis)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.92776    0.04482  -65.33   <2e-16 ***
curr_bmi_num  0.01750    0.00146   11.98   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63470  on 112007  degrees of freedom
AIC: 63474

Number of Fisher Scoring iterations: 5
```

OR shows small increase in risk with BMI, though is statistically significant

[BL Edit - though keep in mind, Odds increased by .02 for each BMI point (e.g. 10 = 20% increase in odds)]

1.6.2 Adjusted, multivariable regression for outcome 1

Regression done again adjusted for variables NOT influenced by BMI (age, sex, ethnicity included - didn't include location as it appears to be the same for all entries)

```
baseline_model <- glm(
hrf_hypercapnic ~ curr_bmi_num + age_at_encounter + sex + ethnicity,
data = stata_hrf_analysis,
family = binomial
)

summary(baseline_model)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_analysis)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -3.3143678  0.0653822 -50.692  < 2e-16 ***
curr_bmi_num                0.0194994  0.0014798  13.177  < 2e-16 ***
age_at_encounter            0.0028333  0.0006493   4.363 1.28e-05 ***
sexMale                     0.2388572  0.0222256  10.747  < 2e-16 ***
ethnicityHispanic or Latino -0.0444911  0.0432380  -1.029    0.303
ethnicityUnknown            0.1622532  0.0302736   5.360 8.34e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63302  on 112003  degrees of freedom
AIC: 63314

Number of Fisher Scoring iterations: 5
```

```r
mods_v1 <- list(
  "HRF v1: unadjusted" = logit_bmi_hrf,
  "HRF v1: adjusted"   = baseline_model
)

modelsummary(
  mods_v1,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```

Association appears to be stronger once adjusted

1.7 Section 6. Sensitivity analyses

1.7.1 Counts for strict HRF definition (v2_strict)

Number of patients meeting strict definition

```r
table(stata_data$hrf_hypercapnic_v2_strict, useNA = "ifany")
```

```
     0      1
817895  15581
```

	HRF v1: unadjusted	HRF v1: adjusted
BMI (per 1 kg/m^2)	1.018	1.020
	[1.015, 1.021]	[1.017, 1.023]
Age (per year)		1.003
		[1.002, 1.004]
Male		1.270
		[1.216, 1.326]
Hispanic/Latino		0.956
		[0.878, 1.040]
Ethnicity: Unknown		1.176
		[1.108, 1.248]
Num.Obs.	112 009	112 009
AIC	63 473.9	63 313.9
BIC	63 493.2	63 371.7
Log.Lik.	−31 734.972	−31 650.965

1.7.2 Unadjusted Logistic Regression for Outcome 2

```r
model_strict_bmi_unadj <- glm(
  hrf_hypercapnic_v2_strict ~ curr_bmi_num,
  family = binomial,
  data = stata_hrf_strict_analysis
)

summary(model_strict_bmi_unadj)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num, family = binomial,
    data = stata_hrf_strict_analysis)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -3.449156   0.059003  -58.46   <2e-16 ***
curr_bmi_num  0.023881   0.001909   12.51   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39644  on 87024  degrees of freedom
Residual deviance: 39491  on 87023  degrees of freedom
  (142832 observations deleted due to missingness)
AIC: 39495

Number of Fisher Scoring iterations: 5
```

1.7.3 Adjusted, Multivariable Regression for Outcome 2

Then adjusted for age, sex, ethnicity:

```r
baseline_model_strict <- glm(
  hrf_hypercapnic_v2_strict ~
    curr_bmi_num +
    age_at_encounter +
    sex +
    ethnicity,
  data = stata_hrf_strict_analysis,
  family = binomial
)

summary(baseline_model_strict)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_strict_analysis)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -4.7689449  0.0935568 -50.974  < 2e-16 ***
curr_bmi_num                0.0277918  0.0019502  14.251  < 2e-16 ***
age_at_encounter            0.0194879  0.0009325  20.900  < 2e-16 ***
sexMale                    -0.0351480  0.0287347  -1.223  0.22126
ethnicityHispanic or Latino -0.1912798  0.0616342  -3.103  0.00191 **
ethnicityUnknown            0.0537704  0.0416200   1.292  0.19638
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39644  on 87024  degrees of freedom
Residual deviance: 38980  on 87019  degrees of freedom
  (142832 observations deleted due to missingness)
AIC: 38992

Number of Fisher Scoring iterations: 6
```

```r
mods_v2 <- list(
  "HRF v2 strict: unadjusted" = model_strict_bmi_unadj,
  "HRF v2 strict: adjusted"   = baseline_model_strict
)

modelsummary(
  mods_v2,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```

```r
mods_all <- list(
  "HRF v1 (primary)"            = baseline_model,
  "HRF v2_strict (sensitivity)" = baseline_model_strict
)

modelsummary(
  mods_all,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

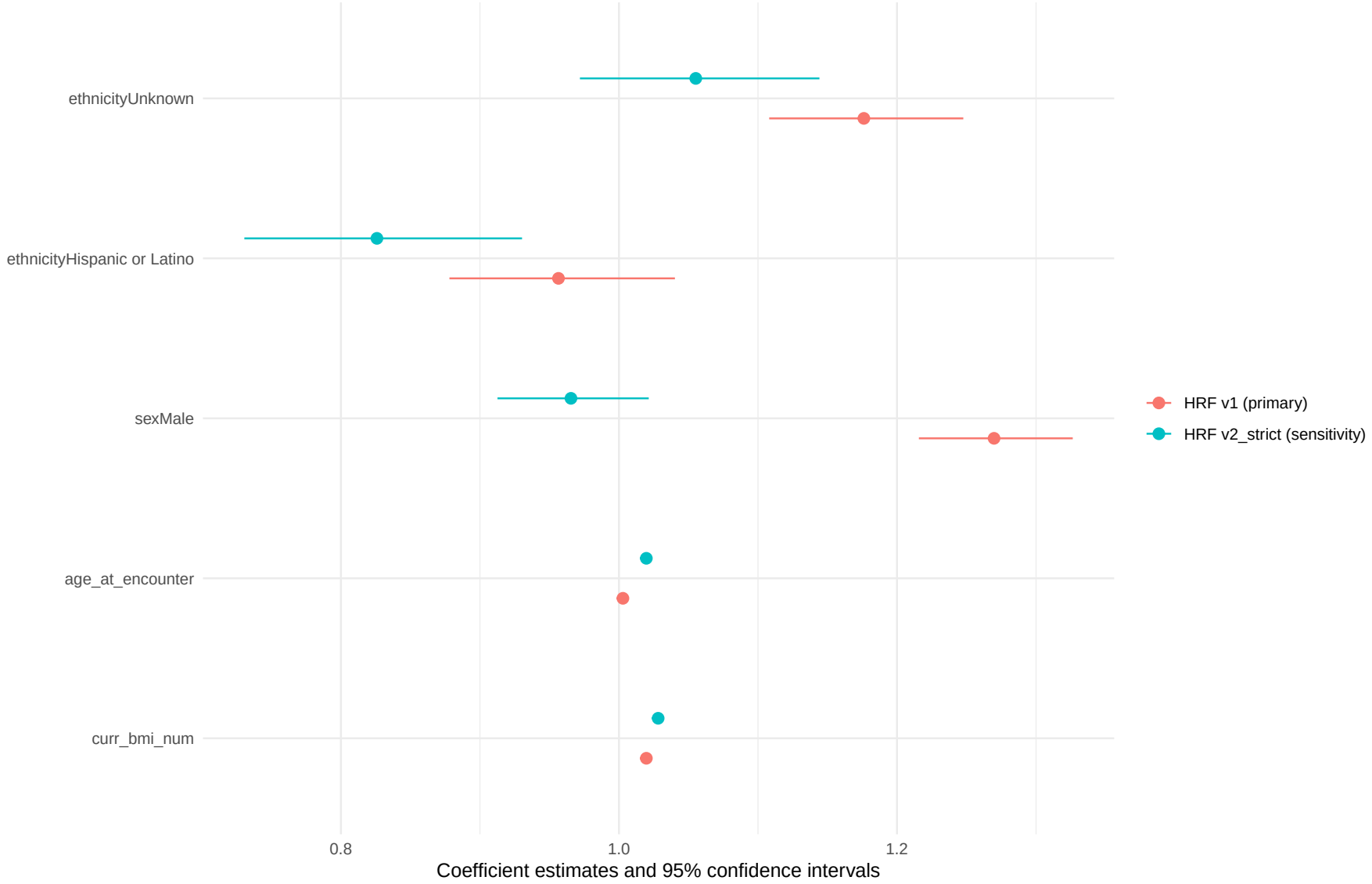
```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```

	HRF v2 strict: unadjusted	HRF v2 strict: adjusted
BMI (per 1 kg/m^2)	1.024	1.028
	[1.020, 1.028]	[1.024, 1.032]
Age (per year)		1.020
		[1.018, 1.022]
Male		0.965
		[0.913, 1.021]
Hispanic/Latino		0.826
		[0.731, 0.930]
Ethnicity: Unknown		1.055
		[0.972, 1.144]
Num.Obs.	87 025	87 025
AIC	39 495.0	38 992.1
BIC	39 513.7	39 048.3
Log.Lik.	−19 745.497	−19 490.036

	HRF v1 (primary)	HRF v2_strict (sensitivity)
BMI (per 1 kg/m^2)	1.020	1.028
	[1.017, 1.023]	[1.024, 1.032]
Age (per year)	1.003	1.020
	[1.002, 1.004]	[1.018, 1.022]
Male	1.270	0.965
	[1.216, 1.326]	[0.913, 1.021]
Hispanic/Latino	0.956	0.826
	[0.878, 1.040]	[0.731, 0.930]
Ethnicity: Unknown	1.176	1.055
	[1.108, 1.248]	[0.972, 1.144]
Num.Obs.	112 009	87 025
AIC	63 313.9	38 992.1
BIC	63 371.7	39 048.3
Log.Lik.	−31 650.965	−19 490.036

```r
modelplot(
  mods_all,
  exponentiate = TRUE,
  coef_omit = "(Intercept)"
)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```


ethnicityUnknown
ethnicityHispanic or Latino
sexMale
age_at_encounter
curr_bmi_num
0.8
1.0
1.2
Coefficient estimates and 95% confidence intervals
HRF v1 (primary)
HRF v2_strict (sensitivity)

1.7.4 Are proportions of hypercapnic respiratory failure (v2) different across BMI categories?

```
# Subset to patients with BMI categories
bmi_hrf_strict_subset <- stata_hrf_strict_analysis %>%
  filter(!is.na(bmi_category))

# Summary table by BMI category
bmi_hrf_strict_summary <- bmi_hrf_strict_subset %>%
  group_by(bmi_category) %>%
  summarise(
    n_total = n(),
    n_hrf = sum(hrf_hypercapnic_v2_strict == 1),
    n_no_hrf = sum(hrf_hypercapnic_v2_strict == 0),
    pct_hrf = round(100 * n_hrf / n_total, 1),
    pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
  ) %>%
  arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Chi-square test
chisq_test_strict <- chisq.test(table(
  bmi_hrf_strict_subset$bmi_category,
  bmi_hrf_strict_subset$hrf_hypercapnic_v2_strict
))

# View summary table and p-value
bmi_hrf_strict_summary
```

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	3866	377	3489	9.8	90.2
Normal weight	24496	1301	23195	5.3	94.7
Overweight	25060	1165	23895	4.6	95.4
Obesity I	16721	906	15815	5.4	94.6
Obesity II	9273	666	8607	7.2	92.8

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Obesity III	7609	832	6777	10.9	89.1

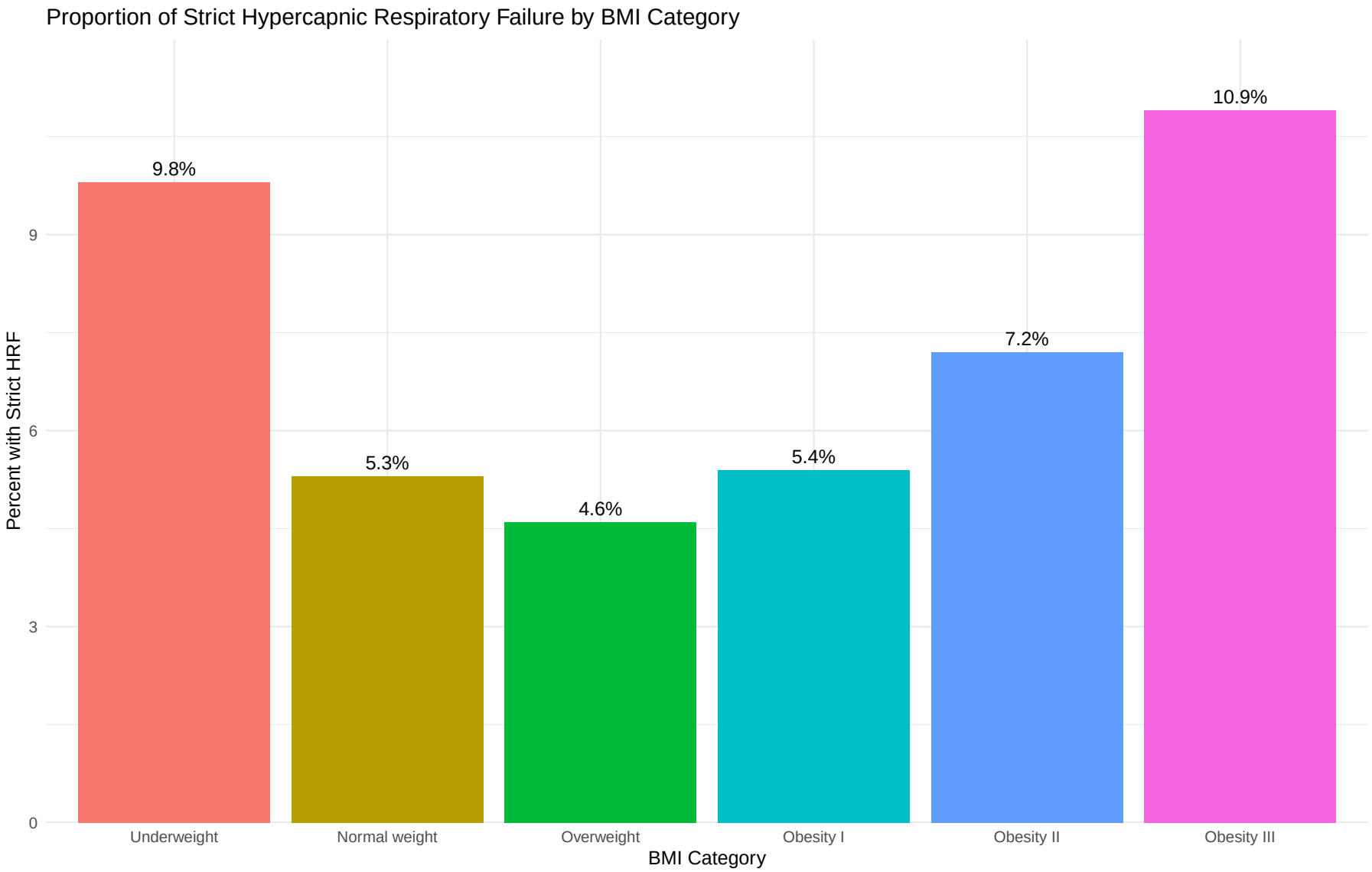
```
chisq_test_strict$p.value
```

```
[1] 3.879776e-118
```

and visualization

```
# Make sure bmi_category is ordered
bmi_hrf_strict_summary <- bmi_hrf_strict_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of strict HRF by BMI category
ggplot(bmi_hrf_strict_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Strict Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with Strict HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



1.8 Section 7. Population attributable fraction

TODO

1.9 Appendix

```
sessionInfo()
```

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/Denver
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] tidyr_1.3.1        dplyr_1.1.4        stringr_1.6.0      gtsummary_2.5.0
[5] here_1.0.2         modelsummary_2.6.0 ggplot2_4.0.1      haven_2.5.5

loaded via a namespace (and not attached):
 [1] gt_1.2.0           generics_0.1.4     xml2_1.5.1         stringi_1.8.7
 [5] hms_1.1.4          digest_0.6.38      magrittr_2.0.4     evaluate_1.0.5
 [9] grid_4.5.2         RColorBrewer_1.1-3 cards_0.7.1        fastmap_1.2.0
[13] rprojroot_2.1.1    jsonlite_2.0.0     backports_1.5.0    purrr_1.2.0
[17] scales_1.4.0       cli_3.6.5          rlang_1.1.6        performance_0.16.0
[21] litedown_0.8       commonmark_2.0.0   tinytable_0.16.0   withr_3.0.2
```

```
[25] yaml_2.3.10         datawizard_1.3.0   tools_4.5.2         checkmate_2.3.4
[29] bayestestR_0.17.0   forcats_1.0.1      vctrs_0.6.5         R6_2.6.1
[33] lifecycle_1.0.4     fs_1.6.6           insight_1.4.6       pkgconfig_2.0.3
[37] pillar_1.11.1       gtable_0.3.6       data.table_1.17.8   glue_1.8.0
[41] xfun_0.54           tibble_3.3.0       tidyselect_1.2.1    parameters_0.28.3
[45] rstudioapi_0.17.1   knitr_1.50         farver_2.1.2        htmltools_0.5.8.1
[49] tables_0.9.33       rmarkdown_2.30     labeling_0.4.3      compiler_4.5.2
[53] S7_0.2.1            markdown_2.0
```